

The sedimentation of flexible filaments: Trajectories, particle clouds and a buckling instability

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Abstract

In this fluid dynamics video we explore an array of different possible dynamics for a flexible filament sedimenting in a viscous fluid. The time-dependent shapes and trajectories of the filament are determined analytically and numerically by balancing viscous, elastic and gravitational forces in a slender-body theory for zero-Reynolds number flows. The dynamics are characterized by a single dimensionless elasto-gravitation number. The video shows the process of filament relaxation and reorientation, the formation of particle clouds, and finally a buckling instability.

Introduction

The shapes and dynamics of elastic filaments in viscous fluids are of critical importance in a wide range of biological and technological processes. However, even the study of relatively simple systems can be difficult, due to the complex interactions between elasticity and nonlocal fluid-structure interactions. In a recent paper, the present authors investigated such a fundamental problem, the sedimentation of a single elastic filament in a viscous fluid [1]. A mathematical model was developed by balancing viscous, elastic, and gravitational forces in a slender-body theory for zero-Reynolds number flows. The dynamics of the filament are characterized by a single dimensionless elasto-gravitation number that compares the elastic forces acting on the filament to the gravitational force:

$$\beta = \frac{\pi E a^4}{4 F_G L^2},$$

where E is the elastic modulus of the filament, a is an average cross-sectional radius, L is the filament length, and F_G is the total gravitational force acting on the body. Numerical and analytical approaches are used to study the filament behavior in a number of situations, with excellent agreement between the two [1]. This video illustrates three prominent features of the filament dynamics, which we now describe.

1. Particle trajectories and reorientation

In the first part of the video, numerical simulations are used to show that sedimentation causes weakly flexible filaments to reorient and eventually reach a terminal settling shape. Two non-interacting filaments (with $\beta = 0.02$) are initially placed at the origin at angles $\pi/4$ and $\pi/64$ with respect to gravity, and are allowed to sediment freely. The instantaneous shape and orientation are tracked along with the position of the center of the filaments, and the time-dependent trajectories and filament shapes are illustrated. Analytical expressions for these quantities have been determined using a multiple-scale asymptotic expansion in the large β regime [1]. We proceed to show the terminal sedimenting shapes over a range of the elasto-gravitation number, β . These shapes are found (and can be shown analytically) to be self-similar over a broad range of the elasto-gravitation number. When β is sufficiently small, large filament deformations lead to a horseshoe-like terminal shape.

2. Particle clouds

The complicated sedimentation trajectories shown in the first part of the video are a consequence of the coupling between the filament shape and its translational and rotational velocities. The individual filament trajectories suggest an interesting possibility for the spreading dynamics for a collection of flexible filaments, which we model as non-interacting in the dilute limit. The second part of the video shows the sedimentation of a collection of filaments, initially at the same location but with different orientations with respect to gravity, with $\beta = 0.1$. In stark contrast to the dynamics of rigid rods, flexible filaments are restricted to a cloud with a width that can be shown to be proportional to the elasto-gravitation number [1].

3. Buckling Instability

Finally, it can be shown that the tension acting on a filament backbone when it sediments along its length is compressive in the filament's leading half. This compression leads to a buckling instability if the filament is sufficiently flexible. A linear stability analysis reveals this buckling threshold, as well as the most unstable wavenumbers and their growth rates as a function of the elasto-gravitation number. In the final part of the video, we show this buckling instability by numerical simulations for three different values of the elasto-gravitation number ($\beta = 0.0005, 0.0001, 0.0000625$). Just as predicted by our analysis, smaller wavelength perturbations are more unstable for smaller β (increased filament flexibility).

References

[1] LI, L., MANIKANTAN, H., SAINTILLAN, D., & SPAGNOLIE, S. E. 2013 The sedimentation of flexible filaments, *J. Fluid Mech.*, to appear (arXiv:1306.4692).